Rakan Eyad

Robotics & Embedded Systems Engineer
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Summary

Mechatronic engineering graduate with hands-on experience in robotics, embedded systems, and control. Built an FPGA-driven companion robot in Verilog, shipped the operator stack for a ROS 2 TurtleBot, built a two-axis vision tracking rig on a Raspberry Pi, and migrated an industrial prototype from a microcontroller to a Raspberry Pi.

Experience

System Engineer — ALBON Nov 2024 — Jan 2026
- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021
- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021
- Supported server system testing and maintenance in data-centre environments.

Featured Projects

Autonomous Target Tracking System — Personal build · hardware
- Designed and built a two-axis vision tracking rig on a Raspberry Pi that holds a moving hand or drone centred in frame, acquiring targets with a pretrained detector.
- Implemented one PID loop per axis with a Kalman filter estimating target velocity, so the mount keeps leading the target through frame dropouts.

Assistive Robot (FPGA) — University build · Verilog
- Built 'RoboDog', a Bluetooth-commanded companion robot, entirely in Verilog on an Altera DE2-115: camera line-following, FFT beat detection, and ultrasonic/IR safety interlocks arbitrated by a four-state FSM (team of 5).
- Closed timing by pipelining the critical path, turning a 32.763 ns violation into 6.46 ns of positive slack; demonstrated all tasks on physical hardware.

Autonomous TurtleBot (ROS2) — University build · ROS2
- Built the operator stack for an autonomous retail TurtleBot3 in ROS 2 and C++: an ImGui control panel with live battery plotting, cart/checkout, and revenue reporting, plus the charging behaviour behind it (team of 5).
- Integrated it with the team's Nav2 navigation and QR-driven inventory server over ROS 2 topics and action servers.

Planetary Rover RL Navigation — Thesis · Isaac Sim
- Developed an Honours thesis (78/100, Distinction) on autonomous rover navigation: an A* global planner coupled to a PPO local policy, trained in NVIDIA Isaac Sim with 64 parallel rovers on Martian terrain.
- Hybrid achieved 99.5% success at 10 m (vs 79.5% standalone-PPO) and 90% vs 2% at 50 m; built the A* command term, terrain importer, reward tuning, and a custom evaluation pipeline.

Skills

Firmware & electronics: C / C++ · Arduino, FPGA · Verilog / SystemVerilog, Raspberry Pi · microcontrollers, Sensor integration & calibration, Power wiring & junction boxes, Assembly

Robotics & simulation: ROS2, Isaac Sim / Isaac Lab, State estimation & path planning, HIL simulation, CAD (Fusion/SW), Mechanism design

Control & signals: Advanced control systems, Signal processing & filtering, State estimation, GNC fundamentals, MATLAB, Radar & antenna coursework

Education

B.E. (Honours), Mechatronic Engineering
University of Sydney · 2022 — 2026
Graduated with First Class Honours